Performance Testing

Date	04 July 2026
Team ID	SWTID-2026-2950
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	k6
Type of Testing	Load Testing
Target Module / API	POST /api/chat
Test Environment	Local Host
Test Date	04 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome

1	Normal Load	10	30 sec	All requests should be processed successfully with response time under 2 seconds.
2	Medium Load	25	30 sec	System should remain stable and respond without errors.
3	High Load	50	60 sec	System should handle increased traffic with acceptable response time and minimal errors.
4	Peak Load	100	30 sec	System should remain available without crashing and maintain acceptable performance.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	2.1 ms	Pass	Excellent response time
2	Response Time (Max)	< 5 seconds	7.73 ms	Pass	Well below 5 seconds
3	Throughput (Req/sec)	-	9.97 req/sec	Pass	Stable throughput
4	Error Rate	< 1%	0.00%	Pass	No failed requests
5	CPU Utilization	< 80%	4.0%	Pass	Low CPU usage
6	Memory Utilization	< 80%	1.7%	Pass	

Step 4: Observations & Analysis

Key Findings:

- The application successfully handled 10 virtual users for 30 seconds.
- Average response time was 2.1 ms, which is well below the target limit.
- Maximum response time was 7.73 ms.
- No request failures were observed (0% error rate).
- CPU and memory utilization remained low, indicating stable system performance.

Bottlenecks Identified:

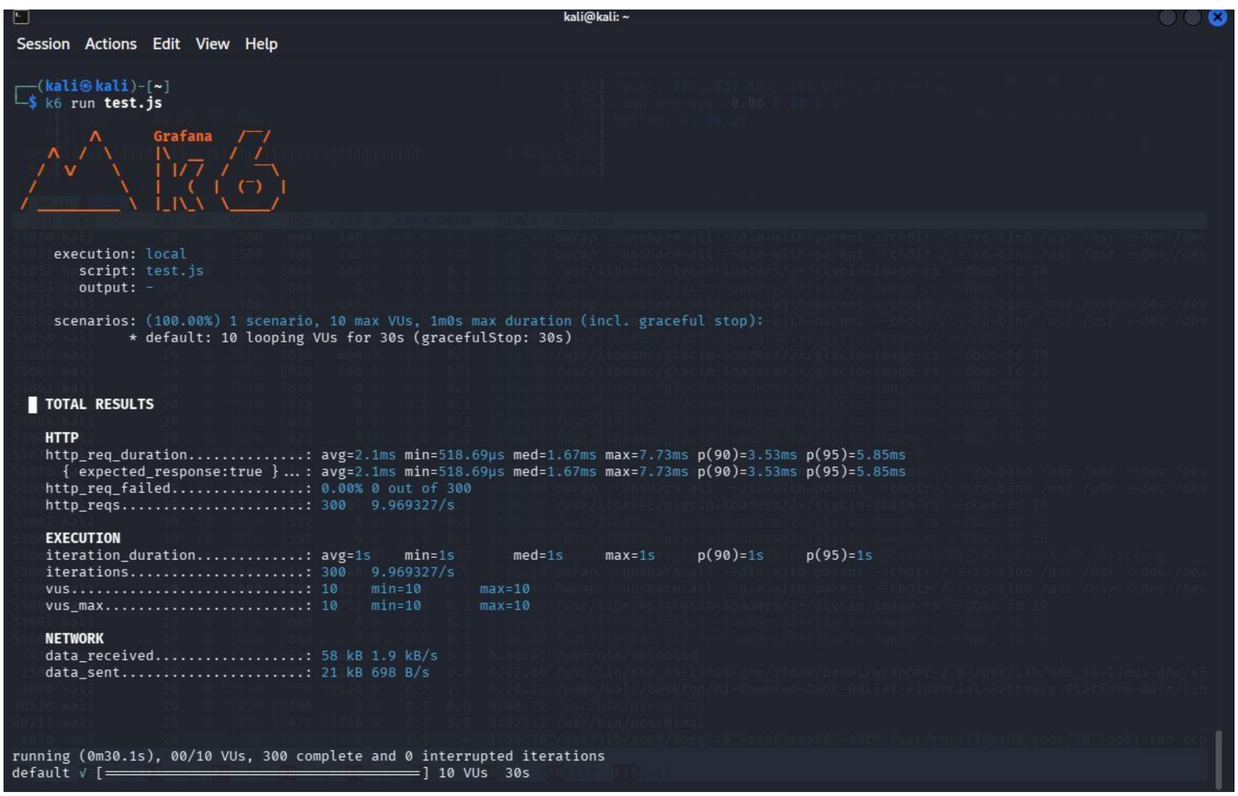
- No major performance bottlenecks were identified during testing.
- The application remained stable under the tested load.
- Response time and resource utilization were within acceptable limits.

Optimization Steps Taken:

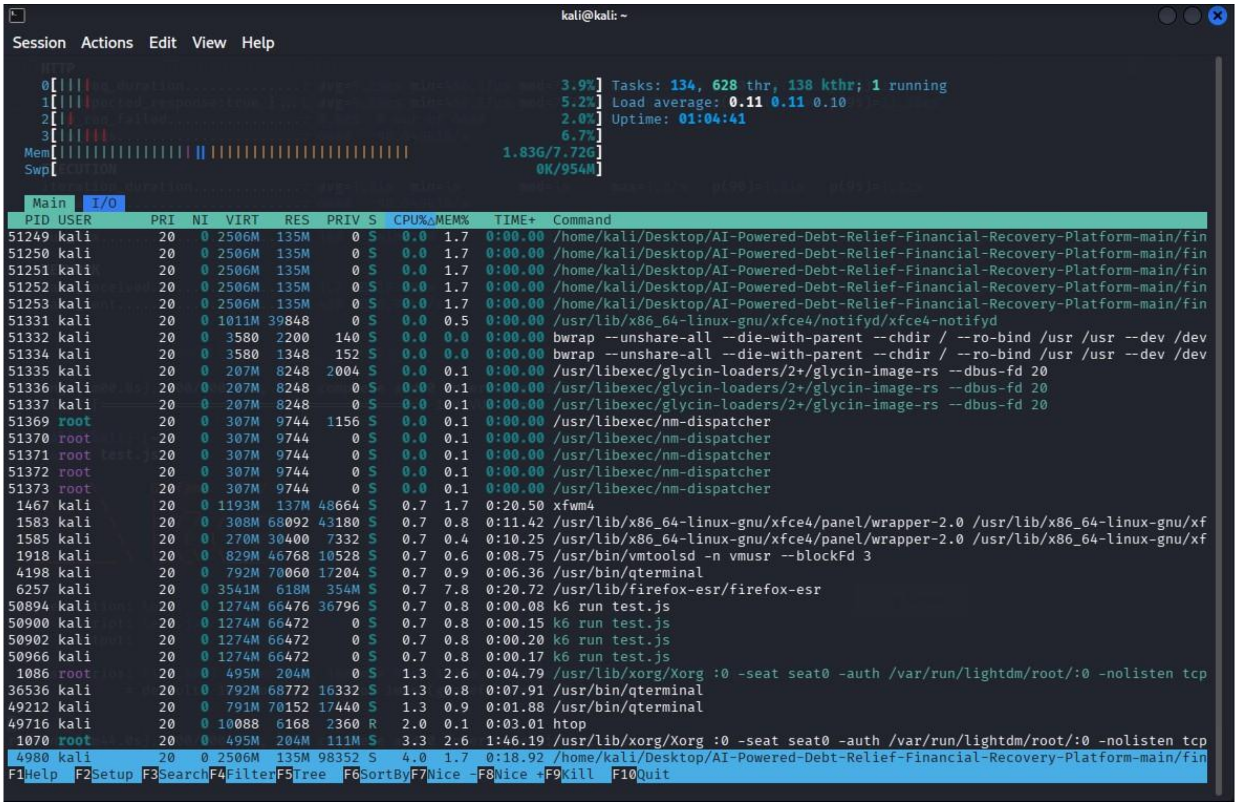
- Verified API performance using k6 load testing.
- Monitored CPU and memory utilization during execution.
- Optimized the test environment by running the application locally.
- No additional optimization was required as the application met the performance targets.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



```
kali@kali: ~
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ k6 run test.js

Grafana

  execution: local
     script: test.js
     output: -

  scenarios: (100.00%) 1 scenario, 10 max VUs, 1m0s max duration (incl. graceful stop):
           * default: 10 looping VUs for 30s (gracefulStop: 30s)

█ TOTAL RESULTS

  HTTP
  http_req_duration.............: avg=2.1ms min=518.69µs med=1.67ms max=7.73ms p(90)=3.53ms p(95)=5.85ms
    { expected_response:true }...: avg=2.1ms min=518.69µs med=1.67ms max=7.73ms p(90)=3.53ms p(95)=5.85ms
  http_req_failed...............: 0.00% 0 out of 300
  http_reqs.....................: 300   9.969327/s

  EXECUTION
  iteration_duration............: avg=1s    min=1s      med=1s     max=1s     p(90)=1s     p(95)=1s
  iterations....................: 300   9.969327/s
  vus...........................: 10    min=10        max=10
  vus_max.......................: 10    min=10        max=10

  NETWORK
  data_received.................: 58 kB 1.9 kB/s
  data_sent.....................: 21 kB 698 B/s

running (0m30.1s), 00/10 VUs, 300 complete and 0 interrupted iterations
default ✓ [======================================] 10 VUs  30s
```



kali@kali: ~
Session Actions Edit View Help
0[|||| 3.9%] Tasks: 134, 628 thr, 138 kthr; 1 running
1[|||| 5.2%] Load average: 0.11 0.11 0.10
2[|| 2.0%] Uptime: 01:04:41
3[|||||| 6.7%]
Mem[|||||||||||||||||||||||||||||||||||||||||| 1.83G/7.72G]
Swp[0K/954M]
Main I/O
PID USER PRI NI VIRT RES PRIV S CPU% MEM% TIME+ Command
51249 kali 20 0 2506M 135M 0 S 0.0 1.7 0:00.00 /home/kali/Desktop/AI-Powered-Debt-Relief-Financial-Recovery-Platform-main/fin
51250 kali 20 0 2506M 135M 0 S 0.0 1.7 0:00.00 /home/kali/Desktop/AI-Powered-Debt-Relief-Financial-Recovery-Platform-main/fin
51251 kali 20 0 2506M 135M 0 S 0.0 1.7 0:00.00 /home/kali/Desktop/AI-Powered-Debt-Relief-Financial-Recovery-Platform-main/fin
51252 kali 20 0 2506M 135M 0 S 0.0 1.7 0:00.00 /home/kali/Desktop/AI-Powered-Debt-Relief-Financial-Recovery-Platform-main/fin
51253 kali 20 0 2506M 135M 0 S 0.0 1.7 0:00.00 /home/kali/Desktop/AI-Powered-Debt-Relief-Financial-Recovery-Platform-main/fin
51331 kali 20 0 1011M 39848 0 S 0.0 0.5 0:00.00 /usr/lib/x86_64-linux-gnu/xfce4/notifyd/xfce4-notifyd
51332 kali 20 0 3580 2200 140 S 0.0 0.0 0:00.00 bwrap --unshare-all --die-with-parent --chdir / --ro-bind /usr /usr --dev /dev
51334 kali 20 0 3580 1348 152 S 0.0 0.0 0:00.00 bwrap --unshare-all --die-with-parent --chdir / --ro-bind /usr /usr --dev /dev
51335 kali 20 0 207M 8248 2004 S 0.0 0.1 0:00.00 /usr/libexec/glycin-loaders/2+/glycin-image-rs --dbus-fd 20
51336 kali 20 0 207M 8248 0 S 0.0 0.1 0:00.00 /usr/libexec/glycin-loaders/2+/glycin-image-rs --dbus-fd 20
51337 kali 20 0 207M 8248 0 S 0.0 0.1 0:00.00 /usr/libexec/glycin-loaders/2+/glycin-image-rs --dbus-fd 20
51369 root 20 0 307M 9744 1156 S 0.0 0.1 0:00.00 /usr/libexec/nm-dispatcher
51370 root 20 0 307M 9744 0 S 0.0 0.1 0:00.00 /usr/libexec/nm-dispatcher
51371 root 20 0 307M 9744 0 S 0.0 0.1 0:00.00 /usr/libexec/nm-dispatcher
51372 root 20 0 307M 9744 0 S 0.0 0.1 0:00.00 /usr/libexec/nm-dispatcher
51373 root 20 0 307M 9744 0 S 0.0 0.1 0:00.00 /usr/libexec/nm-dispatcher
1467 kali 20 0 1193M 137M 48664 S 0.7 1.7 0:20.50 xfwm4
1583 kali 20 0 308M 68092 43180 S 0.7 0.8 0:11.42 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /usr/lib/x86_64-linux-gnu/xf
1585 kali 20 0 270M 30400 7332 S 0.7 0.4 0:10.25 /usr/lib/x86_64-linux-gnu/xfce4/panel/wrapper-2.0 /usr/lib/x86_64-linux-gnu/xf
1918 kali 20 0 829M 46768 10528 S 0.7 0.6 0:08.75 /usr/bin/vmtoolsd -n vmusr --blockFd 3
4198 kali 20 0 792M 70060 17204 S 0.7 0.9 0:06.36 /usr/bin/qterminal
6257 kali 20 0 3541M 618M 354M S 0.7 7.8 0:20.72 /usr/lib/firefox-esr/firefox-esr
50894 kali 20 0 1274M 66476 36796 S 0.7 0.8 0:00.08 k6 run test.js
50900 kali 20 0 1274M 66472 0 S 0.7 0.8 0:00.15 k6 run test.js
50902 kali 20 0 1274M 66472 0 S 0.7 0.8 0:00.20 k6 run test.js
50966 kali 20 0 1274M 66472 0 S 0.7 0.8 0:00.17 k6 run test.js
1086 root 20 0 495M 204M 0 S 1.3 2.6 0:04.79 /usr/lib/xorg/Xorg :0 -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp
36536 kali 20 0 792M 68772 16332 S 1.3 0.8 0:07.91 /usr/bin/qterminal
49212 kali 20 0 791M 70152 17440 S 1.3 0.9 0:01.88 /usr/bin/qterminal
49716 kali 20 0 10088 6168 2360 R 2.0 0.1 0:03.01 htop
1070 root 20 0 495M 204M 111M S 3.3 2.6 1:46.19 /usr/lib/xorg/Xorg :0 -seat seat0 -auth /var/run/lightdm/root/:0 -nolisten tcp
4980 kali 20 0 2506M 135M 98352 S 4.0 1.7 0:18.92 /home/kali/Desktop/AI-Powered-Debt-Relief-Financial-Recovery-Platform-main/fin
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice - F8Nice + F9Kill F10Quit